

Programming Exam 1

Date/Time: 2026.03.10 08:00 – 08:50

(File naming: StudentID_quiz1.py, upload to Moodle PE1 submission area)

Problem: TSMC Stock Price Change Calculator

Develop a small program that allows users to input the "daily percentage change of TSMC's stock price." The input should be a floating-point number (e.g., **1.25** represents a **1.25%** increase, while **-0.80** represents a **0.80%** decrease). Based on this input, the program must calculate and output several intuitive conversion results to help investors quickly grasp the significance of the price movement.

Predefined Settings (Ready to use)

- Assume the "Base Price" is **1,000 TWD per share** (used to calculate today's price).
- Assume you purchased **10 lots** of TSMC stock (1 lot = 1,000 shares), for a total of **10,000 shares**.
- Assume the brokerage fee (for both buying and selling) is **0.1425%** of the total transaction amount.
- Assume the stock transaction tax (applied only upon selling) is **0.3%** of the sale proceeds.

Your program must calculate and output the following four items:

1. Today's Stock Price (TWD/Share)

Calculate the "adjusted stock price" based on the user-inputted percentage change, using the **1,000 TWD benchmark** as the starting point.

2. Change in Stock Position Value (TWD)

Based on a holding of **10 lots (10,000 shares)**, calculate:

- What is the total Stock Position Value after the price change?
- What is the total gain or loss in market value compared to the benchmark value (\$1,000 * 10,000\$)?

3. Equivalent "Per Share Price Change" (TWD/Share)

Although the user inputs a "Percentage Change (%)", investors often find it more intuitive to know the actual dollar amount gained or lost per share. Therefore, you must convert the percentage fluctuation into a "**Per-Share Price Change**." **Example:** For a **1,000 TWD** benchmark price, a **1%** increase equals **+10 TWD** per share.

Parameter Definitions

- P_0 (*Benchmark Stock Price*) : The base price per share (TWD/share); for this problem, it is **1,000**.
- r (*Daily Percentage Change*): The daily price fluctuation (%); this is a floating-point number entered by the user (may be negative).
- P_1 (*Today's Stock Price*) : The calculated price per share after the change (TWD/share).
- ΔP (**Per share price change**): The actual dollar amount increased or decreased per share (TWD/share).

First, calculate Today's Stock Price (convert the percentage into a decimal ratio)

$$: P_1 = P_0 \times \left(1 + \frac{r}{100}\right)$$

Calculate the Per Share Price Change (a positive value indicates an increase; a negative value indicates a decrease) : $\Delta P = P_1 - P_0$

Required Output: :

- ΔP : Calculate the equivalent price change in TWD per share. The output must clearly display the **positive or negative sign** (e.g., **+12.50** or **-8.00**).

4. Minimum Increase to Break Even (Break-even Price Increase rate, %)

Assume you purchased TSMC stock at the **Benchmark Price** and sold it later at a specific "Selling Price." Because trading incurs costs (including **Buying Commission**, **Selling Commission**, and **Securities Transaction Tax**), you may still face a loss even if the stock price remains unchanged. You must calculate: **At what Selling Price (and what equivalent percentage increase) will you exactly break even?**

Parameter Definitions

- P_0 (*Benchmark Stock Price*) : The base price per share (TWD/share); for this problem, it is **1,000**.
- N (*Total Shares*): The total number of shares held; for this problem, it is **10,000**.
- f (*Commission Rate*): The brokerage fee rate (charged for both buying and selling); for this problem, it is **0.1425%** (or **0.001425**).
- t (*Transaction Tax Rate*): The securities transaction tax rate (charged only upon selling); for this problem, it is **0.3%** (or **0.003**).
- P_s (*Break-even Selling Price*): The minimum selling price per share required to cover all costs and achieve zero net profit.

Total Buying Cost (including purchase commission): $\text{BuyCost} = P_0 \times N \times (1 + f)$

Net Selling Revenue (deducting selling commission and transaction tax) : $\text{SellNet} = P_s \times N \times (1 - f - t)$

Break-even Condition (Zero net profit/loss) : $\text{SellNet} = \text{BuyCost}$

by substituting the terms above, solve for : $P_s = \frac{P_0 \times (1+f)}{1-f-t}$

Break-even Price Increaseing Rate : The percentage increase required to cover all costs:

$$: \text{BreakEvenPct} = \left(\frac{P_s}{P_0} - 1\right) \times 100$$

Required Output: :

- P_s : The selling price per share (TWD/share) required to achieve a net profit of zero after all costs.
- BreakEvenPct : The percentage increase (%) needed from the benchmark price to cover all transaction costs.

Input and Output Requirements

- The program must prompt the user to enter the **"Please enter TSMC stock price change rate (%)"**.
- All calculations must be performed using **floating-point numbers (floats)** for precision.
- All results must clearly indicate their respective **units** (e.g., TWD/share, TWD, %).

Sample Input/Output

(The following is the required format and content your must display after execution)

```
C:\Users\user\Desktop\助教\上機考>python pe1_en.py
Please enter TSMC stock price change rate (%) (e.g., 1.25 or -0.8): 1.25

=== TSMC Stock Price Change Calculation Results ===
Reference Stock Price: 1000 dollar/share
Number of Shares: 10 lots (10000 shares)
Today's Price Change Rate: 1.25%

1) Today's Stock Price
   1012.50 yuan/share

2) Changes in Stock Position Value
   Reference Value: 10000000.0 dollar
   Today's Value: 10125000.0 dollar
   Value Change: 125000.0 dollar

3) Equivalent Per-Share Price Change
   Per-Share Change: 12.50 dollar/share

4) Break-even Price Increase Rate (including trading fees and sales tax)
   Trading Fee Rate (Buy/Sell): 0.14250000000000002%
   Sales Tax Rate (Sell): 0.3%
   Break-even Selling Price: 1005.88 dollar/share
   Break-even Price Increase Rate: 0.59%
```

```
C:\Users\user\Desktop\助教\上機考>python pe1_en.py
Please enter TSMC stock price change rate (%) (e.g., 1.25 or -0.8): -0.8

=== TSMC Stock Price Change Calculation Results ===
Reference Stock Price: 1000 dollar/share
Number of Shares: 10 lots (10000 shares)
Today's Price Change Rate: -0.8%

1) Today's Stock Price
   992.00 yuan/share

2) Changes in Stock Position Value
   Reference Value: 10000000.0 dollar
   Today's Value: 9920000.0 dollar
   Value Change: -80000.0 dollar

3) Equivalent Per-Share Price Change
   Per-Share Change: -8.00 dollar/share

4) Break-even Price Increase Rate (including trading fees and sales tax)
   Trading Fee Rate (Buy/Sell): 0.14250000000000002%
   Sales Tax Rate (Sell): 0.3%
   Break-even Selling Price: 1005.88 dollar/share
   Break-even Price Increase Rate: 0.59%
```

Teaching assistants will run your program using **python pe1.py** when grading. After the program starts, it must first pause and display the prompt: **"Please enter TSMC stock price change rate (%) (e.g., 1.25 or -0.80):"** This allows the teaching assistant to input the percentage change. For example, the screenshot above shows the TA entering **1.25**. After the input is provided, your program should then print the calculation results based on the entered percentage. (**Submit the pe1.py file, not a screenshot.**)

Note: You are required to write comments (註解) for each part in your code.